

ML Practical Works (MCA 2nd):

Guideline: To prepare a lab manual for the ML lab works, student should follow a structured, step-by-step format. Each experiment should be documented clearly so students can perform, observe, and conclude.

1. Define Objectives:

- State the purpose of the experiment clearly.
- Example: To implement linear regression using Python.

2. List Requirements:

- Mention tools, libraries, and datasets needed.
- Python version (e.g., 3.10), Anaconda Navigator, VS code
- Libraries: NumPy, Pandas, Matplotlib, Scikit-learn
- Dataset: CSV file or built-in dataset

3. Write Problem Statement:

- Frame the task as a clear question or challenge.
- Example: Predict student pass/fail using logistic regression.

4. Procedure:

- Break down the steps to solve the problem.
- Import libraries
- Load dataset
- Preprocess data
- Apply algorithm
- Evaluate results

5. Include Code:

- Provide clean, well-commented Python code.

6. Show Expected Output:

- Display sample outputs or Screenshots.
- Console output (accuracy, confusion matrix)

- Graphs (scatter plots, regression lines)

7. *Observations:*

- Note what was observed during execution.
- Accuracy achieved
- Behavior of algorithm with different parameters
- Any anomalies or errors

8. *Write Conclusion*

- Summarize findings and learning outcomes.
- State whether objective was achieved
- Mention practical applications
- Suggest improvements or extensions

❖ *Prepare a Lab Manual for the Following Using PYTHON*

Introduction & Libraries

Q₁. Write a Python program to demonstrate basic list, tuple, and dictionary operations.

Q₂. Use NumPy to create a 5×5 matrix and compute its transpose.

Q₃. Data Exploration & Preprocessing

Q₄. Load a CSV dataset using Pandas and display summary statistics.

Q₅. Perform missing value handling (replace with mean/median) on a dataset.

Linear Regression

Q₆. Implement simple linear regression to predict house prices based on square footage.

Q₇. Plot regression line using Matplotlib and interpret the slope and intercept.

Naive Bayes Classifier

Q₈. Implement Naive Bayes classifier on an email dataset to classify spam vs. non-spam.

Q₉. Evaluate accuracy using confusion matrix and classification report.

Logistic Regression

Q₁₀. Implement logistic regression to predict whether a student's passes based on study hours.

Q₁₁. Visualize the decision boundary for logistic regression using Matplotlib.

Decision Trees

Q₁₂. Build a decision tree classifier for the Iris dataset.

Q₁₃. Visualize the decision tree structure using graphviz or sklearn.tree.plot_tree.

Support Vector Machines (SVM)

Q₁₄. Implement SVM with linear kernel on the Iris dataset.

K-Nearest Neighbors (K-NN)

Q₁₅. Implement K-NN classifier to classify handwritten digits (MNIST dataset).

Q₁₆. Experiment with different values of K and observe accuracy changes.

WEKA Applications

Q₁₇. Use WEKA to perform clustering on a dataset (e.g., K-means clustering).

Q₁₈. Compare classification results between WEKA and Python's sklearn for the same dataset.